Basic Transmission Theory

⇒ When a microwave signal transmitted through two-open conductor transmission lines of length larger than the wavelength ($l > \lambda$), energy is lost due to radiation.

⇒ This loss is reduced or eliminated by making transmission of microwave signals through waveguide structures.

⇒ The waveguiding structure may consists of two co-axial conductors or two parallel plates or it may be a single hollow conductor called waveguide.

⇒ The two conductor lines can be analysed in terms of voltage, current and impedance by the
- distributed circuit theory.

⇒ Waveguides can be analysed from the solution of Maxwell's field equations.

⇒ Since the general characteristics of electromagnetic wave propagation in all these lines are same, the basic transmission line theory concering voltage standing waves, reflection and impedance is applicable to problems in all these guiding structures when excited by a single mode.

Transmission line equations

⇒ At fluace frequencies the line parameters are distributed along the whole length of the line in the z-direction.

⇒ A small length of the line can be represented by an equivalent symmetrical T-network with constant parameters R, L, G and C per unit length as shown in fig below:

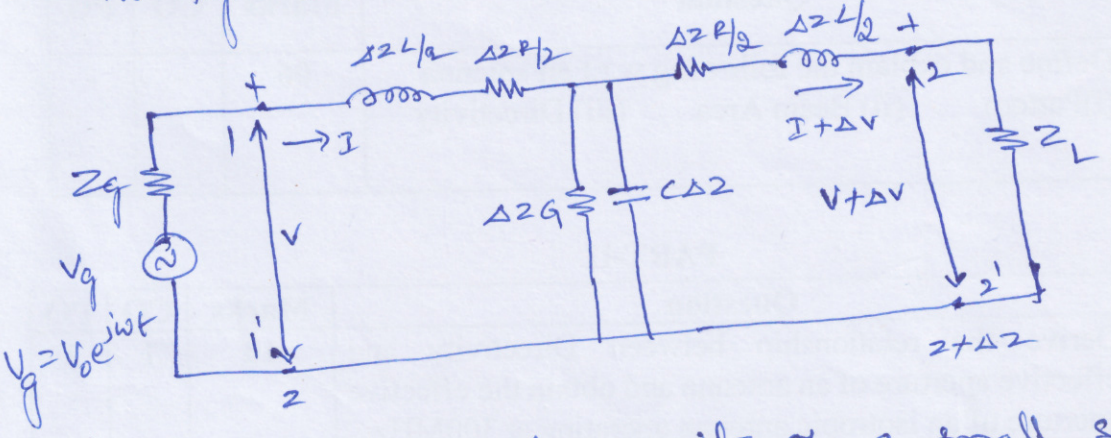


fig: Equivalent circuit of a small section of a uniform transmission line.

Where R → resistance per unit length — accout for ohmic loss

L → inductance per unit length — magnetic energy storage

G → conductance per unit length — dielectric loss bln the line conductors

C → capacitance per unit length — electric energy storage

Assumed that, the line is excited by a matched generator i.e, $Z_g = Z_0$.

In general, the line voltage and current per unit length decreases as 'z' increases, the differential values can be expressed by

i.e., Change in voltage and current,

$$\Delta V = -(R\Delta z + j\omega L \Delta z) I \quad \text{—(1)}$$

$$\Delta I = -(G\Delta z + j\omega C \Delta z) V \quad \text{—(2)}$$

or in the limit $\Delta z \to 0$

i.e., $\lim_{\Delta z \to 0} \frac{\Delta V}{\Delta z} = \frac{dV}{dz} = -(R + j\omega L) I = -ZI$ —(3)

} coupled eqn.

Wy $\lim_{\Delta z \to 0} \frac{\Delta I}{\Delta z} = \frac{dI}{dz} = -(G + j\omega C) V = -YV$ —(4)

} I related to V & vice-versa.

where

$Z = R + j\omega L$

& $Y = G + j\omega C$

} are called the series impedance and shunt admittance per unit length of the line.

Eliminating I from eqn (3) and V from eqn (4), the voltage and current wave equations are generated as

$$\frac{d^2V}{dz^2} = -(R + j\omega L)\frac{dI}{dz}$$

$$= (R + j\omega L)(G + j\omega C) = \gamma^2 V \quad \text{—(5)}$$

Wy $$\frac{d^2I}{dz^2} = -(G + j\omega C)\frac{dV}{dz}$$

$$= (G + j\omega C)(R + j\omega L) I = \gamma^2 I \quad \text{—(6)}$$

Here, $\gamma = \sqrt{(ZY)} = \sqrt{[(R + j\omega L)(G + j\omega C)]}$

$$= \alpha + j\beta. \quad \text{—(7)}$$

and is called the propagation constant.

$\alpha \rightarrow$ attenuation constant

$\beta \rightarrow$ phase constant.

The solution of eqn (5) and (6) gives the voltage and current at any point '2' and are given by

$$V(2) = V_s e^{-\gamma 2} + V_r e^{+\gamma 2} \quad \text{——— (8)}$$

$$I(2) = I_s e^{-\gamma 2} + I_r e^{+\gamma 2} \quad \text{——— (9)}$$

The first term in the right hand side represents the incident wave and the second term represents the reflected wave.

Here,

$V_s \rightarrow$ incident voltage amplitude

$I_s \rightarrow$ incident current amplitude

$V_r \rightarrow$ reflected voltage amplitude

$I_r \rightarrow$ reflected current amplitude.

Some special cases are described below:

(1) Low-Loss line ($R \ll \omega L$, $G \ll \omega C$).

⇒ At high frequencies, when the losses in the line are small,

$R \ll \omega L$, $G \ll \omega C$,

under this condition, eqn (7) can be approximated as

(i) propagation constant

$$\gamma = \alpha + j\beta \simeq j\omega\sqrt{LC}\left[1 + \frac{1}{j2\omega}\left(R/L + G/C\right)\right] m^{-1} \quad \text{——— (10)}$$

by separating real & imaginary parts from eqn (10) we get,

Attenuation constant:

$$\alpha \simeq \frac{1}{2}\left[R\sqrt{C/L} + G\sqrt{L/C}\right] \text{ nepers/m} \quad \text{——— (11)}$$

Phase constant:

$$\beta = \frac{\partial \phi}{\partial z} \simeq \omega\sqrt{LC} \text{ radian/m} \quad \text{—(13)}$$

The velocity of the signal is called the phase velocity and is given by

Phase velocity:

$$v_p = \frac{\partial z}{\partial t} = \omega/\beta \simeq \frac{1}{\sqrt{LC}} = \lambda f \text{ m/sec} \quad \text{—(14)}$$

and is independent of frequency.

The energy of the signal can be considered to be — propagated with group velocity given by

Group velocity:

$$v_g = \frac{\partial \omega}{\partial \beta} = \simeq \frac{c^2}{v_p} \text{ m/sec} \quad \text{—(15)}$$

$c \rightarrow$ velocity of the light in the medium (m/sec).

(11) Lossless line ($R=0$, $G=0$)

⇒ For lines with very good conductors ($\sigma \rightarrow \infty$) and associated with lossless dielectric medium ($\varepsilon''=0$), $R=0=G$.

⇒ From eqn (12) & (13)

$$\alpha = 0$$

$$\gamma = \alpha + j\beta = j\beta \simeq j\omega\sqrt{LC} \text{ m}^{-1} \quad \text{—(16)}$$

$$v_p = \frac{1}{\sqrt{LC}} = \lambda f \text{ m/sec} \quad \text{—(17)}$$

III. Distortion in Transmission Lines

From eqn (7) without any assumption in losses, squaring both sides,

$$\alpha^2 - \beta^2 + j2\alpha\beta = (RG - \omega^2 LC) + j\omega(LG + RC) \quad \text{—(18)}$$

Equating real and imaginary parts from eqn (18)

$$\alpha^2 - \beta^2 = RG - \omega^2 LC \quad \text{—(19)}$$

$$2\alpha\beta = \omega(LG + RC) \quad \text{—(20)}$$

Squaring eqn (20) & solving for α^2

$$4\alpha^2\beta^2 = \omega^2 (LG + RC)^2$$

$$\alpha^2 = \frac{\omega^2 (LG + RC)^2}{4\beta^2}$$

Substitute α^2 in to (19), will get

$$\beta^4 + (RG - \omega^2 LC)\beta^2 - \omega^2/4 (LG + RC)^2 = 0 \quad \text{—(21)}$$

Obtaining the solution for β from eqn (20) and substitute in (18), we get

$$\beta = \left[\frac{\omega^2 LC - RG + \sqrt{(RG - \omega^2 LC)^2 + \omega^2 (LG + RC)^2}}{2} \right]^{1/2} \quad \text{—(22)}$$

$$\alpha = \left[\frac{RG - \omega^2 LC + \sqrt{(RG - \omega^2 LC)^2 + \omega^2 (LG + RC)^2}}{2} \right]^{1/2} \quad \text{—(23)}$$

From eqn (22) & (23), attenuation is a function of frequency.

∴ The signal consisting of a band of frequencies will attenuate differently at different frequency. and the line will introduce frequency distortion.

⇒ The phase constant is not a linear function of frequency but the phase velocity $v_p = \omega/\beta$ of different frequency component of the signal will have the diffent velocities and signal will suffer "Dispersion", which is a kind of distortion called "Delay distortion"

∴ The net result will be the distortion of the signal at the receiver end.

IV. Distortion less lines

⇒ A line is said to be distortion free when α and $v_p = \omega/\beta$ are both independent of frequency & thus 'β' is a direct function of frequency.

For loss less lines, $R = G = 0$, then Eqn (22) & (23)

$$\alpha = 0 \quad \text{and} \quad \beta = \omega\sqrt{LC}.$$

$$v_p = \omega/\beta = 1/\sqrt{LC} = \text{constant}$$

and β is a linear function of frequency.

∴ "loss less lines are distortion free"

⇒ For lossy lines,

In case $R \neq 0$, $G \neq 0$, then if the inner square root term in eqn (22) becomes $(RG + \omega^2 LC)^2$

or $\left[(RG - \omega^2 LC)^2 + \omega^2 (LG + RC)^2\right]^{1/2} = \left[(RG + \omega^2 LC)^2\right]^{1/2}$

Squaring both sides and equating coefficients of corresponding powers of 'ω' on both sides of the eqn, we get,

$$(LG)^2 + (RC)^2 - 2RGLC = 0$$

$$\text{or } (LG - RC)^2 = 0$$

$$\text{or } \boxed{R/L = G/C} \quad \text{—(24)}$$

"This is the condition of no distortion in the line"

Substitute this condition to eqn (22) & (23),

$\alpha = \sqrt{RG} = R\sqrt{C/L}$, independent of frequency

$\beta = \omega\sqrt{LC}$ = linear function of frequency

$V_p = 1/\sqrt{LC}$ = constant

$V_g = 1/\sqrt{LC}$ = constant.

∴ eqn (24) is the condition for distortionless line.